Seasonal Demand Forecasting and Return Behavior Analysis in Cross-Border E-Commerce

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Abstract

Accurate demand forecasting and return management are important for cross-border e-commerce. Sales often fluctuate sharply across weekdays and shopping seasons, while returns and cancellations create additional pressure after orders are completed. Existing retail forecasting work mainly focuses on sales prediction, but pays less attention to how return behavior evolves after high-sales periods and how this affects operations. This report uses the Online Retail II transaction dataset to construct daily and weekly time series, aiming to compare revenue forecasting models and examine the relationship among revenue, orders, customers, and return or cancellation behavior. After separating normal sales from returns and cancellations, we apply baseline models, ARIMA, SARIMA, Prophet, DHR-ARIMA, SARIMAX, and weekly VAR analysis, and evaluate forecasting performance under fixed-origin, rolling hold-out, and rolling-origin cross-validation schemes. The forecasting results do not identify a single universally best model: performance changes with the evaluation window and error metric. SARIMA is more robust for the final year-end hold-out period, especially in fixed-origin forecasting and rolling MAE/RMSE, while Seasonal Naive remains highly competitive and performs best in rolling sMAPE. In the rolling-origin CV period, Prophet full achieves the lowest MAE and RMSE, indicating that deterministic seasonality can work well in smoother validation windows but is less robust around the final year-end demand peak. The VAR results provide little evidence of strong lagged cross-variable effects on revenue growth after differencing and system controls. Return analysis shows that high-revenue weeks are mainly associated with higher return or cancellation volumes in the same week and the following week, with weaker evidence of a longer response. Overall, the study highlights the value of simple seasonal benchmarks and suggests that returns should be monitored as a separate post-sales process.

Keywords: Time Series Analysis, Demand Forecasting, Seasonal Forecasting, SARIMA, VAR Model, Return Behavior, Cross-Border E-Commerce.

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1 Introduction

1.1 Research Background

Cross-border e-commerce has become an important part of modern retailing. Compared with traditional offline retail, online retailers face more frequent demand fluctuations, stronger calendar effects, and more visible return or cancellation behavior. For this reason, sales forecasting is not only a statistical exercise, but also a practical tool for inventory planning, logistics scheduling, and after-sales resource allocation. Retail forecasting has long been recognized as a key operational problem in both research and practice (Fildes et al., 2022), and large-scale retail forecasting studies such as the M5 competition further show the value of comparing forecasting models on real sales data (Makridakis et al., 2022).

At the same time, returns and cancellations are also important in e-commerce operations. They may occur after sales peaks and create additional pressure on inventory, customer service, and cash-flow management. Previous studies have shown that return policies and return behavior are closely related to consumer purchase and return decisions (Janakiraman et al., 2016), while recent work has also highlighted e-commerce return forecasting as a growing research topic (Karl, 2025). Motivated by these issues, this study uses the Online Retail II transaction dataset to construct daily and weekly time series, and analyzes both sales demand forecasting and return/cancellation dynamics.

1.2 Research Questions

This study focuses on the following questions:

- What trend, volatility, and calendar patterns can be observed in daily and weekly sales revenue?
- Which forecasting models perform better for daily revenue under fixed-origin, rolling hold-out, and rolling-origin cross-validation settings?
- Do weekly orders, customers, and return/cancellation orders provide meaningful dynamic information beyond revenue's own history?
- Is return or cancellation pressure mainly concentrated immediately after high-revenue weeks, or does it show a longer lagged response?

1.3 Structure and Contributions

The rest of this report is organized as follows. Section 2 describes the Online Retail II dataset, data cleaning rules, and time series construction. Section 3 presents exploratory analysis of sales seasonality, operational variables, return/cancellation behavior, and basic diagnostics. Section 4 introduces the three empirical modules: daily revenue forecasting, weekly VAR analysis, and return-lag analysis. Section 5 reports the main findings, while Section 6 discusses their operational implications and limitations. Section 7 concludes the report.

This report makes several contributions. First, it transforms raw transaction records into interpretable daily and weekly time series by separating normal sales from returns, cancellations, and other non-sales records. Second, it compares multiple forecasting models, including baseline models, ARIMA, SARIMA, Prophet, DHR-ARIMA, and SARIMAX, under fixed-origin, rolling hold-out, and rolling-origin cross-validation

settings, showing that model performance depends on both the evaluation window and the error metric. Third, it uses weekly VAR analysis to examine whether orders, customers, and return/cancellation orders provide dynamic information beyond revenue's own history. Finally, it treats return and cancellation behavior as a separate post-sales process and examines when return pressure appears after high-revenue periods.

2 Data Source and Preprocessing

2.1 Data Source and Variable Description

The data used in this study are the Online Retail II dataset from Kaggle Mashlyn (2020), with the original version also available from the UCI Machine Learning Repository UCI Machine Learning Repository (2019). The dataset contains transaction-level records from a UK-based online retailer between 1 December 2009 and 9 December 2011.

Each transaction record includes invoice information, product code and description, quantity, invoice time, unit price, customer identifier, and country. These fields allow us to distinguish normal sales from cancellations or returns, compute transaction-level revenue, and aggregate the raw records into daily and weekly time series. A detailed description of the original variables is reported in Appendix A.

2.2 Data Cleaning Rules and Results

The purpose of data cleaning is to separate normal sales from returns, cancellations, and other non-sales records, so that the constructed revenue series has a clear economic interpretation. The following cleaning rules are applied:

- Fully duplicated rows are removed.
- Records with $Quantity_i > 0$, $Price_i > 0$, and non-cancellation invoices are treated as normal sales.
- Records with $Quantity_i < 0$ or cancellation invoices are treated as returns or cancellations; all remaining records are classified as other records.
- Revenue is computed only for normal sales records, while missing Customer ID values are ignored only when counting unique customers.

Table 1 reports the main cleaning results.

Table 1: Summary of Data Cleaning Results

Category	Count	Percentage
Raw records	1,067,371	–
Fully duplicated records	34,335	3.22% of raw records
Records after de-duplication	1,033,036	100.00%
Normal sales records	1,007,913	97.57%
Return/cancellation records	22,497	2.18%
Other records	2,626	0.25%
Normal sales invoices	40,077	–
Return/cancellation invoices	11,685	–
Normal sales revenue	20,476,260.45	–

After de-duplication, missing Customer ID values, missing product descriptions, and non-positive prices remain in the data; these records are handled according to their analytical roles rather than being removed.

2.3 Time Series Variable Construction

Since the original data are transaction-level records, they are aggregated into daily and weekly time series before modeling. For period t, let $\mathcal{S}_t$ denote the set of normal sales records and $\mathcal{C}_t$ denote the set of return or cancellation records. The main aggregated variables are defined as

$$\begin{aligned} Revenue_t &= \sum_{i \in \mathcal{S}_t} Quantity_i \times Price_i, \\ Orders_t &= \#\{Invoice_i : i \in \mathcal{S}_t\}, \\ Customers_t &= \#\{CustomerID_i : i \in \mathcal{S}_t,\ CustomerID_i \neq \text{NA}\}, \\ ReturnOrders_t &= \#\{Invoice_i : i \in \mathcal{C}_t\}. \end{aligned} \tag{1}$$

Daily series are used for forecasting models, while weekly series are used for VAR analysis to reduce high-frequency noise. Calendar dates without normal sales records are retained and assigned zero revenue and zero order count. Among the 739 calendar days, 604 days contain normal sales orders and 135 days have no normal sales records. Log-transformed variables, such as $\log(1 + Revenue_t)$, are also constructed for later modeling. Table 2 summarizes the main constructed time series datasets.

Table 2: Constructed Time Series Datasets

Dataset	Frequency	Period Covered	Observations
Daily retail time series	Daily	2009-12-01 to 2011-12-09	739
Weekly retail time series	Weekly	2009-11-30 to 2011-12-05	106
VAR candidate series	Weekly, differenced	After first differencing	105

The aggregation is internally consistent: total revenue, total normal sales invoices, and total return/-cancellation invoices match the corresponding cleaning records in Table 1.

3 Exploratory Data Analysis

3.1 Sales Trends and Seasonal Patterns

Figure 1 shows weekly revenue and its 4-week moving average. Weekly aggregation reduces daily noise and makes the year-end pattern clearer. Revenue rises sharply near the end of both 2010 and 2011, while the beginning of each year shows relatively low sales. The increase in late 2011 is very strong, suggesting a clear shopping-season effect. The first and last weeks are partial weeks, so they should be interpreted cautiously.

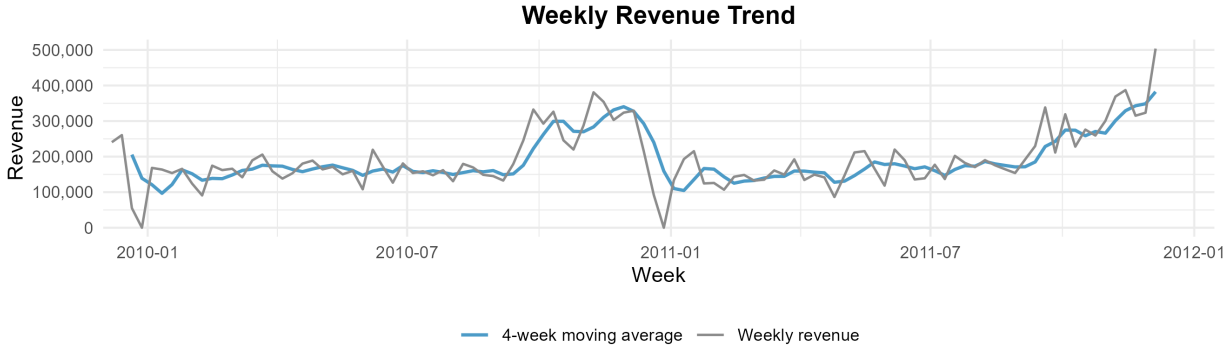


Figure 1: Weekly Revenue Trend

Figure 2 further summarizes average daily revenue by weekday and month. The weekday pattern is clear: revenue is much higher from Monday to Friday, especially on Tuesday and Thursday, while Saturday sales are almost zero. The monthly pattern also shows strong seasonality, with average daily revenue increasing from September to November and peaking in November.

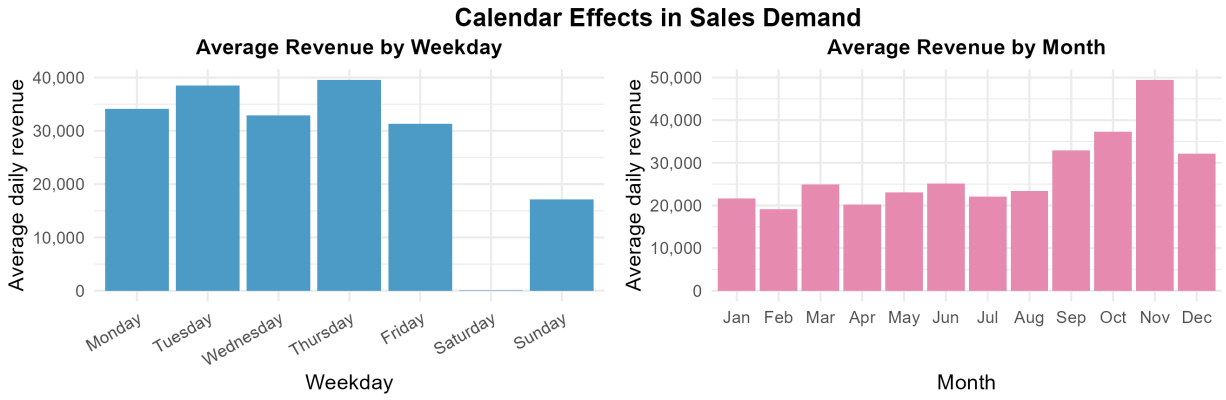


Figure 2: Calendar Effects in Sales Demand

These patterns show that sales demand is not purely irregular. It contains clear weekly and yearly seasonality, together with noticeable volatility around high-demand periods. This motivates the later use of seasonal forecasting models such as SARIMA, deterministic trend-seasonality models such as Prophet, and SARIMAX models with calendar-related covariates.

3.2 Key Operational Variables and Preliminary Dependence

Figure 3 compares the weekly operational variables used in the VAR analysis. Panel (a) shows the standardized series of revenue, orders, customers, and return/cancellation orders. Revenue, orders, and customers move closely together, especially during high-demand periods. Return/cancellation orders fluctuate more sharply and are less aligned with the sales-side variables, suggesting that returns may follow a different post-sales pattern.

Panel (b) reports the contemporaneous Pearson correlations. Revenue is highly correlated with orders and customers, with coefficients of 0.889 and 0.888, respectively. Orders and customers are almost perfectly correlated at 0.989, so they mainly reflect the same sales activity. Return/cancellation orders are also positively related to the sales-side variables, but the correlations are weaker, ranging from 0.387 to 0.494.

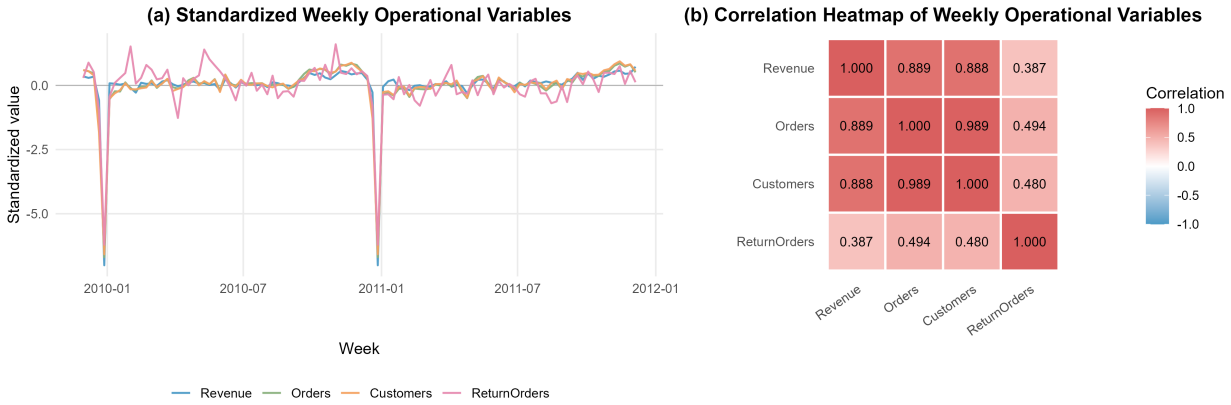


Figure 3: Key weekly operational variables. Panel (a) plots the standardized weekly series of revenue, orders, customers, and return/cancellation orders. Panel (b) reports their contemporaneous Pearson correlations.

These patterns provide a useful starting point for the VAR analysis. Orders and customers are used to measure sales-side activity, while return/cancellation orders capture after-sales pressure. However, the correlations in Figure 3 are contemporaneous, so they should not be read as evidence of predictive causality. Lagged relationships are examined later through Granger causality tests, impulse responses, and forecast error variance decomposition.

3.3 Return and Cancellation Pattern Analysis

The weekly return rate averages about 30%, indicating a heavy return burden. This is consistent with the view that product returns are not merely accounting reversals, but an important part of the firm–customer exchange process (Petersen and Kumar, 2009). Panel (a) shows a mild decline after the elevated period in early 2010, but no year-end surge matching the revenue peak. Panel (d) shows monthly variation: median return rates exceed 30% in January, February, and May, and fall to about 20–25% from August to November. The January–February pattern is consistent with post-holiday returns. The May peak may reflect seasonal factors that are not observed in the dataset.

We then test whether returns trail sales. The CCF is computed between $\Delta \log(\text{Weekly Revenue})$ and $\Delta \log(\text{Weekly Return Orders})$. Panel (c) shows one significant positive correlation at $k = 4$ weeks. The coefficient is about 0.49 and exceeds the 95% bounds, while negative lags are insignificant. This suggests that, in the differenced log series, return orders lag sales by about four weeks. Panel (b) shows a similar timing pattern. This result motivates Section 5.3, with one caveat: in levels, the link is strongest in the same-week and next-week windows.



Figure 4: Return and cancellation patterns. Panel (a): return rate and 4-week moving average; panel (b): standardized revenue and return orders; panel (c): CCF with 95% confidence bounds; panel (d): return rates by month.

3.4 Pre-estimation Time Series Diagnostics

Table 3 reports the ADF tests. Weekly revenue, orders, and customers have unit roots in levels, but their first differences are stationary at the 1% level. We therefore treat them as $I(1)$ series. Weekly return orders are level-stationary, suggesting short-lived return shocks. Daily log revenue rejects the level unit-root null ($p = 0.022$); however, its ACF decays slowly and shows seasonal spikes. We use $D = 1$ in the SARIMA model to align the specification with seasonal differencing.

Table 3: ADF unit-root tests. Lag length is selected by AIC. †: $p < 0.05$; ‡: $p \leq 0.01$; Δ: first difference.

Series	Transform	ADF Statistic	p-value	Conclusion
log(Daily Revenue)	Level	-3.738	0.022	Stationary†
	Δ	-11.336	< 0.01	Stationary‡
log(Weekly Revenue)	Level	-1.557	0.760	Unit root
	Δ	-6.374	< 0.01	Stationary‡
log(Weekly Orders)	Level	-2.431	0.398	Unit root
	Δ	-5.816	< 0.01	Stationary‡
log(Weekly Customers)	Level	-2.687	0.292	Unit root
	Δ	-6.408	< 0.01	Stationary‡
log(Weekly Return Orders)	Level	-4.301	0.010	Stationary‡
	Δ	-6.136	< 0.01	Stationary‡

Figure 5 shows the ACF and PACF of daily log revenue. In levels, the ACF decays slowly and spikes near multiples of lag 7, confirming weekly seasonality ($s = 7$). After differencing, correlations fall sharply, but seasonal spikes remain at lags 7 and 14. The PACF has a clear negative spike at lag 1, followed by rapid decay. These patterns motivate SARIMA specifications of the form $(p, d, q) \times (P, 1, Q)_7$, where $p, q, P, Q \in \{0, 1\}$.

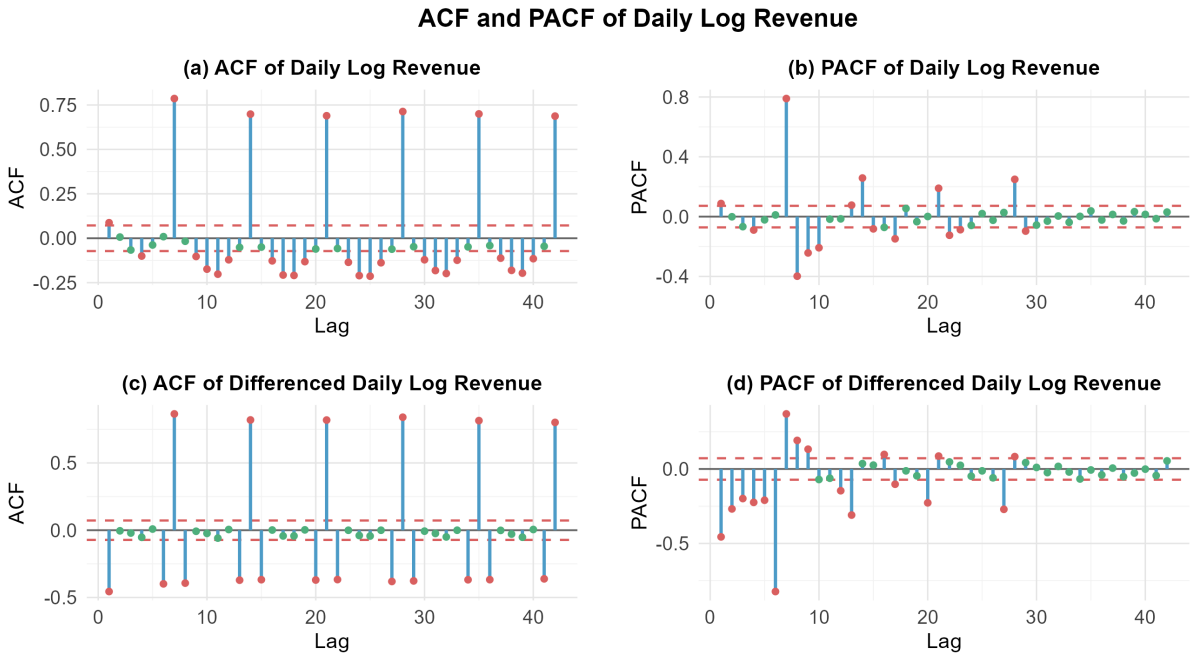


Figure 5: ACF and PACF of daily log revenue in levels and first differences.

Figure 6 shows the ACFs of the differenced weekly series used in the VAR model. Most short-lag autocorrelations lie within the confidence bounds, so the transformed series are close to white noise. A few isolated spikes indicate modest short-run dependence, which a low-order VAR such as VAR(1) should capture.

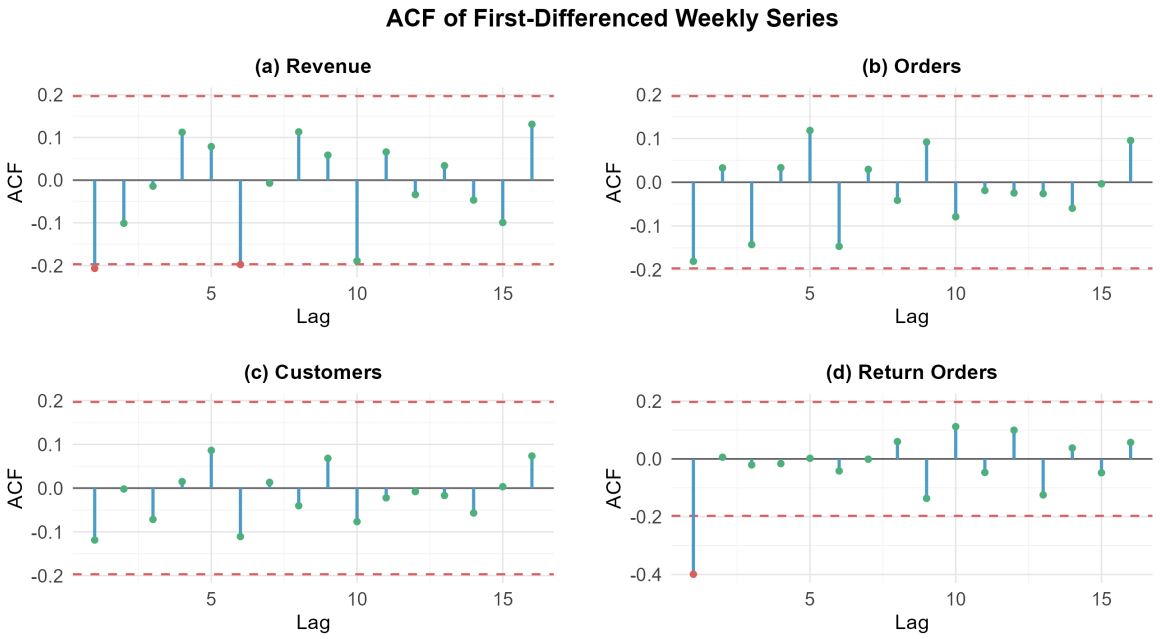


Figure 6: ACF of the four differenced weekly series used in the VAR model, up to lag 16.

4 Methodology

This section introduces three empirical modules. The first compares daily revenue forecasting models under fixed-origin, rolling hold-out, and rolling-origin cross-validation schemes. The second uses a weekly VAR model to study the dynamic links among revenue, orders, customers, and return/cancellation orders. The third examines whether high-sales weeks are followed by delayed return or cancellation pressure. Together, these modules move from sales forecasting to multivariate dynamics and then to post-sales timing. Table 4 summarizes the role of each method in this empirical design.

Table 4: Overview of Empirical Modules and Model Roles

Module	Method	Role
Daily revenue forecasting	Naive, Seasonal Naive ARIMA, SARIMA Prophet, DHR-ARIMA SARIMAX	Benchmark persistence and weekly recurrence. Model stochastic dependence and weekly seasonality. Capture deterministic trend and seasonal patterns. Test calendar and lagged operational predictors.
Weekly dynamic analysis	VAR Granger causality, IRF, FEVD	Model lagged interactions among weekly variables. Assess predictive links, shock responses, and variance shares.
Return-lag analysis	Forward correlation and abnormal peak matching	Check return pressure after high-sales weeks.

4.1 Experiment I: Daily Revenue Forecasting

4.1.1 Forecasting Target and Evaluation Schemes

The first experiment evaluates out-of-sample forecasts of daily revenue. Let R_t denote daily sales revenue, namely $Revenue_t$ at the daily frequency. The forecasting target is the log-transformed series

$$X_t = \log(1 + R_t). \tag{2}$$

Forecasts are transformed back to the original revenue scale by

$$\widehat{R}_t = \exp(\widehat{X}_t) - 1. \tag{3}$$

Any negative revenue forecasts after back-transformation are truncated at zero.

We use three evaluation schemes. The fixed-origin scheme trains each model once and forecasts the final hold-out period. The one-step-ahead rolling hold-out scheme updates the information set before each forecast date, making it closer to daily operational forecasting. Rolling-origin cross-validation is conducted before the final hold-out period as a robustness check.

Because these schemes use different information sets and evaluation periods, model comparisons are made within each scheme. We do not choose a single global best model across all settings.

4.1.2 Model Hierarchy for Forecasting

All forecasting models use the transformed daily revenue series $X_t = \log(1 + R_t)$. We group them by information type to examine whether seasonality, deterministic trends, or operational predictors improve on simple benchmarks. Detailed implementation settings, including fixed model orders and automatic selection rules, are summarized in Appendix B.1; the main text focuses on the modeling logic.

Baseline recurrence models. We first use two simple benchmarks. Let $\widehat{X}_T(h)$ denote the forecast of X_{T+h} based on information available at time T. The naive forecast uses the latest observed value:

$$\widehat{X}_T(h) = X_T, \qquad h = 1, \ldots, H. \tag{4}$$

The seasonal naive forecast uses the most recent value from the same seasonal position. With weekly seasonality $s = 7$,

$$\widehat{X}_T(h) = X_{T+h-ms}, \qquad m = \left\lceil \frac{h}{s} \right\rceil, \quad s = 7. \tag{5}$$

In one-step-ahead rolling forecasts, this uses the value from seven days earlier. The two benchmarks capture persistence and weekly recurrence.

Autoregressive seasonal models. We next estimate ARIMA and SARIMA models for X_t. Let B be the backshift operator, with $BX_t = X_{t-1}$. A non-seasonal ARIMA(p, d, q) model is written as

$$\phi(B)(1 - B)^d X_t = \theta(B)\varepsilon_t, \tag{6}$$

where $\phi(B)$ and $\theta(B)$ are the autoregressive and moving-average polynomials, and $\{\varepsilon_t\}$ is white noise. Intercept or drift terms are included when needed. In the fixed-origin comparison, ARIMA and SARIMA orders are selected using the automatic ARIMA procedure implemented in the `forecast` package, which combines differencing decisions and information-criterion-based order selection (Hyndman and Khandakar, 2008). Given the clear weekly pattern in daily revenue, we also estimate SARIMA models with $s = 7$:

$$\phi(B)\Phi(B^s)(1 - B)^d(1 - B^s)^D X_t = \theta(B)\Theta(B^s)\varepsilon_t, \qquad s = 7, \tag{7}$$

where $\Phi(B^s)$ and $\Theta(B^s)$ are the seasonal autoregressive and moving-average polynomials. ARIMA serves as a non-seasonal reference, while SARIMA is the main univariate seasonal model.

Deterministic seasonality models. Prophet and DHR-ARIMA are used to test whether deterministic seasonal components improve forecasts. Following the decomposable forecasting framework of Taylor and

Letham (2018), Prophet represents the transformed series as

$$X_t = g(t) + s(t) + \varepsilon_t, \tag{8}$$

where $g(t)$ captures trend and $s(t)$ captures weekly and yearly seasonality.

DHR-ARIMA uses Fourier terms for weekly and annual seasonality, following the idea of representing complex seasonal patterns through trigonometric terms (De Livera et al., 2011):

$$X_t = \beta_0 + \boldsymbol{\beta}'\mathbf{F}_t + n_t, \tag{9}$$

where $\mathbf{F}_t$ contains weekly and annual sine-cosine Fourier terms with selected orders K_w and K_y, and n_t follows an ARIMA error process. These models assess whether flexible deterministic seasonality adds value beyond weekly recurrence and SARIMA dynamics.

Dynamic regression with operational predictors. Finally, SARIMAX adds external predictors. Let $\mathbf{Z}_t$ denote these regressors. The model is

$$X_t = \beta_0 + \boldsymbol{\beta}'\mathbf{Z}_t + u_t, \tag{10}$$

where $\mathbf{Z}_t$ includes calendar variables and lagged operational predictors, such as lagged orders and lagged return/cancellation orders. Same-day operational variables are excluded to reduce information leakage. The rolling SARIMAX evaluation is the main setting for interpreting operational predictors under a realistic one-step-ahead information set. The error term u_t follows a seasonal ARIMA process:

$$\phi(B)\Phi(B^s)(1-B)^d(1-B^s)^D u_t = \theta(B)\Theta(B^s)\varepsilon_t, \qquad s = 7. \tag{11}$$

4.1.3 Forecast Accuracy Measures and Residual Diagnostics

Forecast accuracy is evaluated on the original revenue scale using MAE, RMSE, and sMAPE. For N forecast observations,

$$\begin{aligned} \text{MAE} &= \frac{1}{N}\sum_{t=1}^{N}|R_t - \widehat{R}_t|, \\ \text{RMSE} &= \left\{\frac{1}{N}\sum_{t=1}^{N}(R_t - \widehat{R}_t)^2\right\}^{1/2}, \\ \text{sMAPE} &= \frac{100}{N}\sum_{t=1}^{N}\frac{2|R_t - \widehat{R}_t|}{|R_t| + |\widehat{R}_t|}. \end{aligned} \tag{12}$$

When both R_t and $\widehat{R}_t$ are zero, the corresponding sMAPE term is set to zero.

We also use Ljung–Box tests to check whether forecast residuals $e_t^{(X)} = X_t - \widehat{X}_t$ contain serial dependence:

$$Q(m) = N(N+2)\sum_{k=1}^{m}\frac{\widehat{\rho}_k^2}{N-k}, \tag{13}$$

where $\widehat{\rho}_k$ is the residual autocorrelation at lag k. The null hypothesis is that residual autocorrelations up to lag m are jointly zero. We set $m = 14$ to check for remaining two-week dependence in daily forecast errors.

4.2 Experiment II: Weekly VAR Dynamic Estimation

Experiment II studies weekly interactions among revenue, orders, customers, and return/cancellation orders. Unlike the daily forecasting task, this analysis focuses on system dynamics and structural interpreta-

tion. A VAR is suitable because it treats all variables as endogenous and models their lagged interdependence, following the standard multivariate time-series framework introduced by Sims (1980).

4.2.1 Variable Construction and Stationarity Treatment

Let the raw weekly vector be

$$\mathbf{W}_t = \big(Revenue_t,\ Orders_t,\ Customers_t,\ ReturnOrders_t\big)^\top. \tag{14}$$

The VAR input is defined as

$$\mathbf{x}_t = \Delta \log(\mathbf{1} + \mathbf{W}_t). \tag{15}$$

with the transformation applied element-wise. This reduces scale differences and focuses the analysis on weekly changes.

ADF tests in Section 3.4 show that the four transformed series are stationary. The log-level of $ReturnOrders_t$ is already stationary, so differencing may be excessive in theory. It is still retained for consistency across variables. The first and last partial weeks are removed. Four Christmas–New Year shutdown weeks are also excluded because their absolute log-differences exceed five and mainly reflect temporary closure.

4.2.2 VAR Specification, Lag Selection, and Diagnostics

For the stationary vector $\mathbf{x}_t$, the VAR(p) model is

$$\mathbf{x}_t = \boldsymbol{\phi}_0 + \boldsymbol{\Phi}_1 \mathbf{x}_{t-1} + \cdots + \boldsymbol{\Phi}_p \mathbf{x}_{t-p} + \boldsymbol{\varepsilon}_t, \qquad \boldsymbol{\varepsilon}_t \sim WN(\mathbf{0}, \boldsymbol{\Sigma}_\varepsilon). \tag{16}$$

Here, $\boldsymbol{\phi}_0$ is the intercept vector, and $\boldsymbol{\Phi}_1, \ldots, \boldsymbol{\Phi}_p$ are coefficient matrices. Lag order is selected using AIC, BIC/SC, and HQ. Given the limited weekly sample, BIC/SC is prioritised when the criteria disagree. Model adequacy is checked using characteristic-root moduli and a multivariate Portmanteau test.

4.2.3 Dynamic Interpretation

The fitted VAR is interpreted through Granger causality tests, impulse response functions (IRFs), and forecast error variance decomposition (FEVD).

Pairwise Granger causality, in the sense of whether lagged values of one variable improve the prediction of another, is tested with the joint F-test (Granger, 1969). For target variable x_i and cause variable x_j, the null hypothesis is

$$H_0 :\ \phi_{ij,1} = \cdots = \phi_{ij,p} = 0. \tag{17}$$

Rejecting H_0 means that lags of x_j help predict x_i, conditional on the lagged history of the full VAR system.

Orthogonalised IRFs are computed using the Cholesky decomposition

$$\boldsymbol{\varepsilon}_t = P\boldsymbol{\eta}_t, \qquad \mathbb{E}(\boldsymbol{\eta}_t \boldsymbol{\eta}_t^\top) = I. \tag{18}$$

where P is the lower-triangular Cholesky factor of $\boldsymbol{\Sigma}_\varepsilon$. The ordering is

$$Revenue \ \rightarrow\ Orders \ \rightarrow\ Customers \ \rightarrow\ ReturnOrders.$$

Revenue is ordered first, so its structural shock absorbs contemporaneous correlations with later variables. Orders and customers are treated as same-week operational outcomes, while returns are assumed to adjust with delay. This ordering affects only the orthogonalised IRFs and FEVD, not the reduced-form VAR or

Granger causality tests.

IRFs are computed over a 12-week horizon with 200 bootstrap replications and 95% confidence bounds. FEVD reports the share of forecast-error variance explained by each structural shock at horizon h:

$$\theta_{ij}(h) = \frac{\sum_{r=0}^{h-1} \left(\mathbf{e}_i^\top \boldsymbol{\Psi}_r P \mathbf{e}_j\right)^2}{\sum_{r=0}^{h-1} \mathbf{e}_i^\top \boldsymbol{\Psi}_r \boldsymbol{\Sigma}_\varepsilon \boldsymbol{\Psi}_r^\top \mathbf{e}_i}. \tag{19}$$

where $\mathbf{e}_i, \mathbf{e}_j$ are unit selection vectors and $\boldsymbol{\Psi}_r$ is the r-step moving-average coefficient matrix.

4.3 Experiment III: Return Lag and Sales Anomaly Detection

Experiment III examines whether unusually high sales weeks are followed by delayed return/cancellation pressure. Let S_t denote weekly revenue, and let B_t denote weekly return/cancellation orders.

We first compute forward Pearson correlations:

$$\widehat{\rho}_{SB}(k) = \mathrm{Corr}(S_t,\, B_{t+k}), \qquad k = 0, 1, \ldots, 4. \tag{20}$$

These correlations use raw weekly levels. Because Pearson correlation is scale-invariant, this is equivalent to computing $\mathrm{Corr}(Z_t^S, Z_{t+k}^B)$, where Z^S and Z^B are the standardised series. The four-week horizon reflects a typical online retail return cycle. This level-based check complements the CCF analysis in Section 3.3, which uses first-differenced log series.

Abnormal peaks are identified from

$$Z_t^S = \frac{S_t - \bar{S}}{s_S}, \qquad Z_t^B = \frac{B_t - \bar{B}}{s_B}. \tag{21}$$

Week t is labelled as an abnormal revenue peak if $Z_t^S > \tau$, and as an abnormal return-order peak if $Z_t^B > \tau$, with $\tau = 2$. Under a normal approximation, this threshold selects about the top 2.3% of observations. For each revenue peak, we check whether a return-order peak occurs within $t+1, \ldots, t+4$. This analysis describes timing patterns rather than structural causality.

5 Empirical Results

The previous sections described the data construction and exploratory patterns. Daily revenue shows strong weekly and year-end seasonality, weekly revenue moves closely with orders and customer activity, and return/cancellation behavior shows additional dynamics. Based on these findings, this section reports the results of the three empirical experiments: daily revenue forecasting, weekly VAR-based dynamic analysis, and return-lag and abnormal peak analysis.

5.1 Experiment I: Daily Revenue Forecasting Results

This experiment evaluates daily revenue forecasts under three settings: fixed-origin forecasting, one-step-ahead rolling hold-out forecasting, and rolling-origin cross-validation. Because these settings use different information sets and evaluation periods, models are compared within each setting rather than through a single overall ranking, following the general forecasting practice that model performance should be evaluated under clearly defined out-of-sample settings (Petropoulos et al., 2022).

5.1.1 Fixed-Origin Forecasting Results

Table 5 reports fixed-origin results for the final 56-day hold-out period. SARIMA performs best, with the lowest MAE, RMSE, and sMAPE. Seasonal Naive is very close to SARIMA, showing that weekly recurrence is already a strong forecasting signal. By contrast, ARIMA performs poorly, especially in sMAPE, suggesting that ignoring weekly seasonality leads to large errors.

DHR-ARIMA and Prophet full both improve on ARIMA, but neither outperforms SARIMA or Seasonal Naive. This indicates that deterministic seasonality alone does not fully capture the final hold-out period, especially around the year-end demand window. Prophet full is therefore better viewed as a deterministic trend-seasonality benchmark in this setting.

Table 5: Fixed-Origin Forecast Accuracy and Residual Diagnostics

Model	MAE	RMSE	sMAPE (%)	Ljung-Box p-value
SARIMA	19,254	30,695	39	0.040
Seasonal Naive	19,501	31,254	40	< 0.001
Naive	26,140	35,335	68	< 0.001
DHR-ARIMA	29,732	44,856	81	< 0.001
Prophet full	33,461	49,198	83	< 0.001
ARIMA	46,130	55,778	172	< 0.001

Figure 7 compares actual revenue with the fixed-origin forecasts. SARIMA and Seasonal Naive follow the weekly recurrence pattern more closely than the other models. ARIMA underestimates most nonzero sales days, while Prophet full and DHR-ARIMA deviate more during the year-end demand period. These visual patterns are consistent with the accuracy results in Table 5.

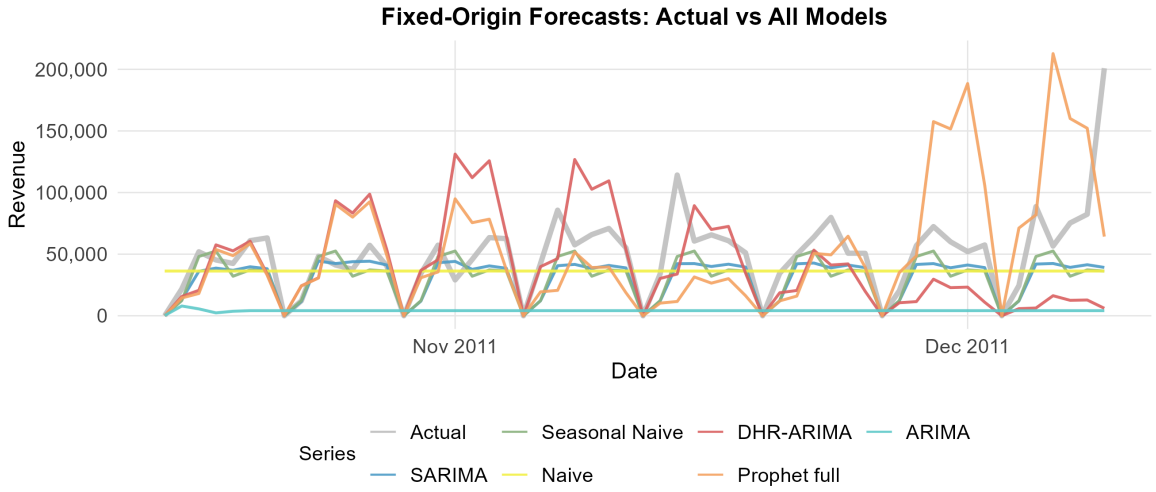


Figure 7: Fixed-origin forecasts: actual revenue versus all models.

Residual diagnostics lead to the same conclusion. ARIMA residuals retain clear serial dependence, while SARIMA substantially reduces the seasonal residual pattern. Although SARIMA residuals are not fully white noise, the results show that adding weekly seasonality improves both forecast accuracy and residual behavior.

5.1.2 Rolling Hold-Out Forecasting Results

Table 6 reports the one-step-ahead rolling hold-out results. Rolling SARIMA achieves the lowest MAE and RMSE, while Rolling Seasonal Naive has the lowest sMAPE. This suggests that SARIMA improves absolute-error accuracy, but weekly recurrence remains highly effective in relative-error terms.

Rolling Lagged-Only SARIMAX is also competitive, with MAE 13,932 and RMSE 24,622, though it does not outperform Rolling SARIMA. Lagged operational variables therefore add some short-term information, but their gains are limited once the information set is updated each day. Rolling DHR-ARIMA and Rolling Prophet full perform worse, indicating that their deterministic seasonal structures fit the final year-end hold-out period less well than SARIMA or the weekly recurrence benchmark.

Table 6: Rolling Hold-Out Forecast Accuracy and Residual Diagnostics

Model	MAE	RMSE	sMAPE (%)	Ljung-Box p-value
Rolling SARIMA	12,836	23,599	36	0.059
Rolling Seasonal Naive	13,501	25,001	24	0.076
Rolling Lagged-Only SARIMAX	13,932	24,622	42	0.636
Rolling DHR-ARIMA	26,763	40,738	74	< 0.001
Rolling Prophet full	30,012	44,908	78	< 0.001

Figure 8 compares actual revenue with the rolling forecasts. Rolling SARIMA, Rolling Seasonal Naive, and Rolling Lagged-Only SARIMAX follow the short-run weekly pattern more closely, while Rolling Prophet full and Rolling DHR-ARIMA deviate more around the year-end demand period. Overall, rolling updates improve SARIMA's practical forecasting performance, but the weekly benchmark remains competitive.

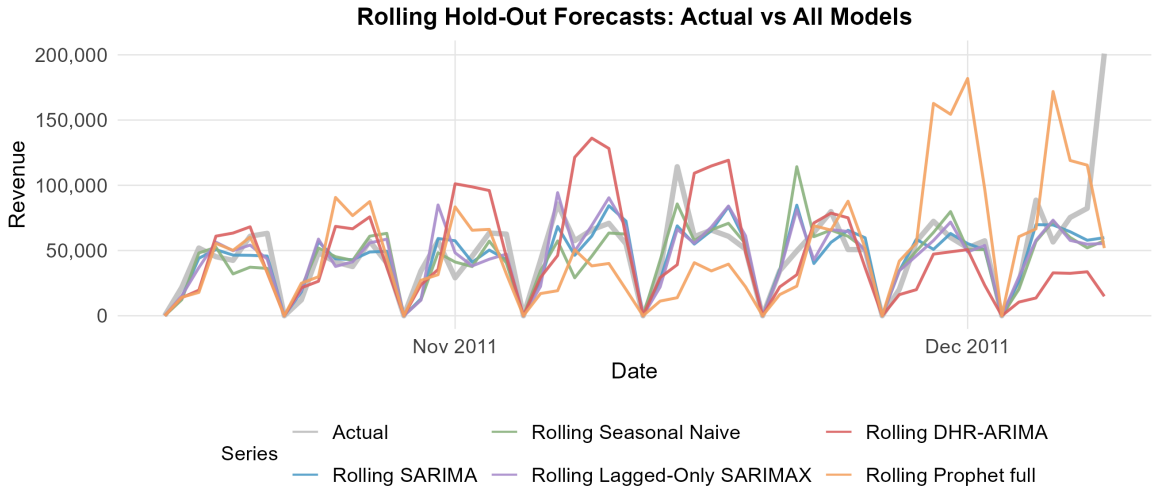


Figure 8: Rolling hold-out forecasts: actual revenue versus all models.

Table 7 further compares SARIMAX specifications. The lagged-only specifications use lagged orders and lagged return/cancellation orders, while the other specifications rely on pre-known calendar or event indicators.

The lagged-only SARIMAX specifications perform best among the SARIMAX variants, while calendar-only and event-window specifications do not improve accuracy. This suggests that lagged operational variables are more informative than simple event indicators, although the comparison should be interpreted under a rolling operational information set.

Table 7: Forecast Accuracy of SARIMAX Specifications

Model	MAE	RMSE	sMAPE (%)
Lagged-Only-SARIMAX-fixed	14,918	26,915	41
Lagged-Only-SARIMAX-auto	14,930	26,927	41
Original SARIMAX	25,436	40,960	85
Calendar-SARIMAX-auto	28,666	43,077	94
Hybrid-Calendar-Event-SARIMAX-auto	33,679	52,335	79
Event-Window-SARIMAX-auto	38,835	55,032	91

5.1.3 Rolling-Origin Cross-Validation and Robustness

Rolling-origin cross-validation checks whether the final rolling hold-out results are stable in earlier periods. This time-ordered validation design is consistent with the view that time series forecasting should preserve the temporal structure of the data during evaluation (Bergmeir et al., 2018). Table 8 shows that Rolling Prophet full CV achieves the lowest MAE and RMSE, followed by Rolling DHR-ARIMA CV and Rolling Seasonal Naive CV. Rolling Seasonal Naive CV still has the lowest sMAPE, confirming that weekly recurrence remains a strong relative-error benchmark.

The ranking differs from the final rolling hold-out results. Prophet full performs well in the CV period, but this advantage does not carry over to the final hold-out window. Deterministic trend-seasonality is therefore useful in some validation periods, but less robust around the year-end demand peak.

Table 8: Rolling-Origin Cross-Validation Forecast Accuracy

Model	MAE	RMSE	sMAPE (%)	Ljung-Box p-value
Rolling Prophet full CV	11,613	16,321	66	0.991
Rolling DHR-ARIMA CV	12,136	17,187	71	0.284
Rolling Seasonal Naive CV	12,317	17,543	40	0.044
Rolling SARIMA CV	17,281	28,212	69	0.047
Rolling Lagged-Only SARIMAX CV	18,539	31,331	69	0.012

Figure 9 compares actual revenue with the rolling-origin CV forecasts. The CV period has a different demand pattern from the final hold-out period, so the results should be read as a robustness check rather than a substitute for the final test. Rolling Prophet full CV and Rolling DHR-ARIMA CV reduce absolute forecast errors in this period, while Rolling Seasonal Naive CV remains strong under sMAPE.

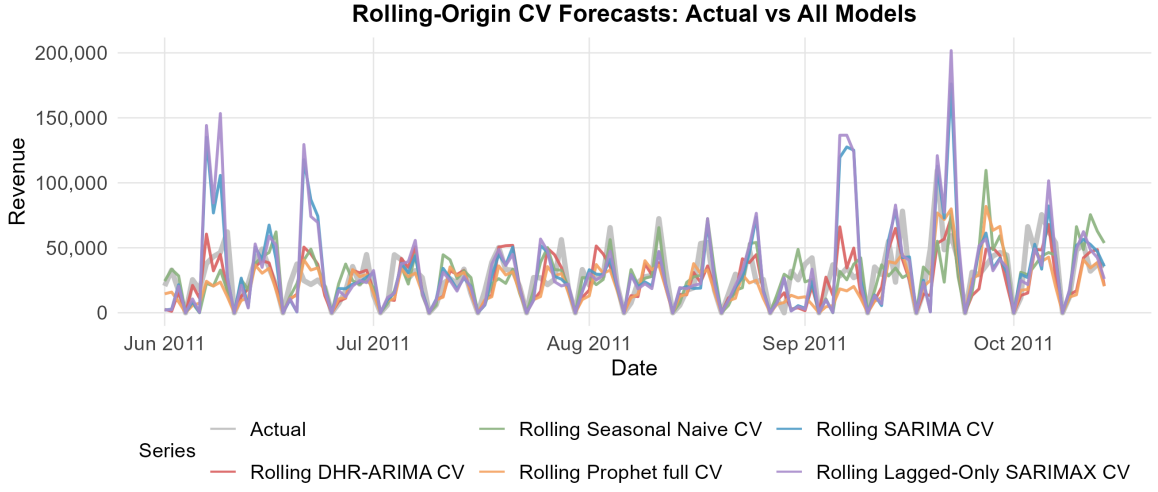


Figure 9: Rolling-origin CV forecasts: actual revenue versus all models.

5.1.4 Forecast Diagnostics and Summary

Residual diagnostics further support the forecasting comparison. In the fixed-origin setting, most models have small Ljung–Box p-values, indicating remaining serial dependence in the forecast residuals. This is expected because the final 56-day hold-out period is forecast without updating the information set and is close to the year-end demand peak.

Rolling updates improve residual behavior. Rolling SARIMA, Rolling Seasonal Naive, and Rolling Lagged-Only SARIMAX have Ljung–Box p-values close to or above the 5% level, while Rolling DHR-ARIMA and Rolling Prophet full show clear residual dependence. This suggests that rolling updates, weekly recurrence, and lagged operational predictors help reduce systematic forecast errors in the final hold-out period.

Table 9 summarizes the best models under each evaluation setting. SARIMA performs best in fixed-origin forecasting, and Rolling SARIMA gives the lowest MAE and RMSE in rolling hold-out. Rolling Seasonal Naive performs best under sMAPE in both rolling evaluations. In rolling-origin CV, Rolling Prophet full CV has the lowest MAE and RMSE, but this advantage does not carry over to the final hold-out window.

Table 9: Best Forecasting Models by Evaluation Setting

Setting	Best MAE	Best RMSE	Best sMAPE
Fixed-origin	SARIMA	SARIMA	SARIMA
Rolling hold-out	Rolling SARIMA	Rolling SARIMA	Rolling Seasonal Naive
Rolling CV	Rolling Prophet full CV	Rolling Prophet full CV	Rolling Seasonal Naive CV

Overall, no model dominates across all settings and metrics. SARIMA is more robust for the final year-end hold-out period, while Seasonal Naive remains a strong weekly benchmark. Prophet full and DHR-ARIMA perform better in the smoother rolling-origin CV period, but their advantage weakens in the final hold-out window. Lagged-Only SARIMAX provides limited gains from operational predictors.

5.2 Experiment II: VAR Dynamic Estimation Results

5.2.1 Model Setup and Diagnostics

The VAR system uses four first-differenced log weekly variables: $\Delta \log(1+Revenue_t)$, $\Delta \log(1+Orders_t)$, $\Delta \log(1+Customers_t)$, and $\Delta \log(1+ReturnOrders_t)$. The sample follows the restrictions in Section 4. The BIC/SC criterion selects a VAR(1), consistent with the weekly ACF in Figure 6.

Table 10 summarizes the diagnostics. The system is stable, with a largest characteristic root modulus of 0.4623, and the Portmanteau test is not significant ($p = 0.3186$), suggesting no remaining residual serial correlation. Table 11 reports the VAR(1) coefficients. Because weekly orders and customers are highly collinear, individual reduced-form coefficients should not be overinterpreted; the main directional evidence is therefore taken from the pairwise Granger tests.

Table 10: Diagnostic results for the estimated VAR(1) model.

Diagnostic	Result	Interpretation
Selected lag order	$p = 1$	Selected by BIC/SC priority
Maximum lag considered	8	Weekly lag search range
Characteristic root moduli	0.4623, 0.3463, 0.3463, 0.0678	All below one
Portmanteau statistic	184.31	Residual serial correlation test
Portmanteau p-value	0.3186	No significant residual autocorrelation

Table 11: Estimated VAR(1) coefficient matrix. Rows are dependent variables; columns are lagged predictors. Standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	$\Delta \log \text{Rev}_{t-1}$	$\Delta \log \text{Ord}_{t-1}$	$\Delta \log \text{Cus}_{t-1}$	$\Delta \log \text{Ret}_{t-1}$
$\Delta \log \text{Rev}_t$	-1.381^{***}	-12.253^{***}	15.236^{***}	0.318
	(0.395)	(1.949)	(1.444)	(0.300)
$\Delta \log \text{Ord}_t$	-0.470^{**}	-6.130^{***}	7.108^{***}	0.206
	(0.193)	(0.952)	(0.705)	(0.146)
$\Delta \log \text{Cus}_t$	-0.398^{**}	-5.617^{***}	6.414^{***}	0.174
	(0.190)	(0.940)	(0.697)	(0.145)
$\Delta \log \text{Ret}_t$	-0.326^{*}	-4.902^{***}	6.012^{***}	-0.340^{**}
	(0.179)	(0.884)	(0.655)	(0.136)
Constant	0.004	-0.002	-0.001	-0.006
	(0.117)	(0.057)	(0.056)	(0.053)

5.2.2 Granger Causality Results

Figure 10 reports the pairwise Granger causality tests. Each test asks whether one lagged variable improves prediction of a target variable, after controlling for all lagged variables in the VAR. This is different from a system-level `vars::causality()` test, which tests whether one variable jointly predicts all other equations.

The results are conservative. Orders do not Granger-cause revenue ($p = 0.995$), and neither do customers ($p = 0.726$) or return orders ($p = 0.823$). Revenue also does not Granger-cause return orders ($p = 0.786$). Thus, the VAR(1) provides no statistically significant one-week predictive direction among the main relationships of interest.

This does not contradict the CCF and forward-correlation results reported elsewhere. Those analyses use different transformations and lag structures. The CCF captures growth-rate co-movement over several weeks, whereas the VAR test focuses on one-step-ahead prediction conditional on the full lagged system.

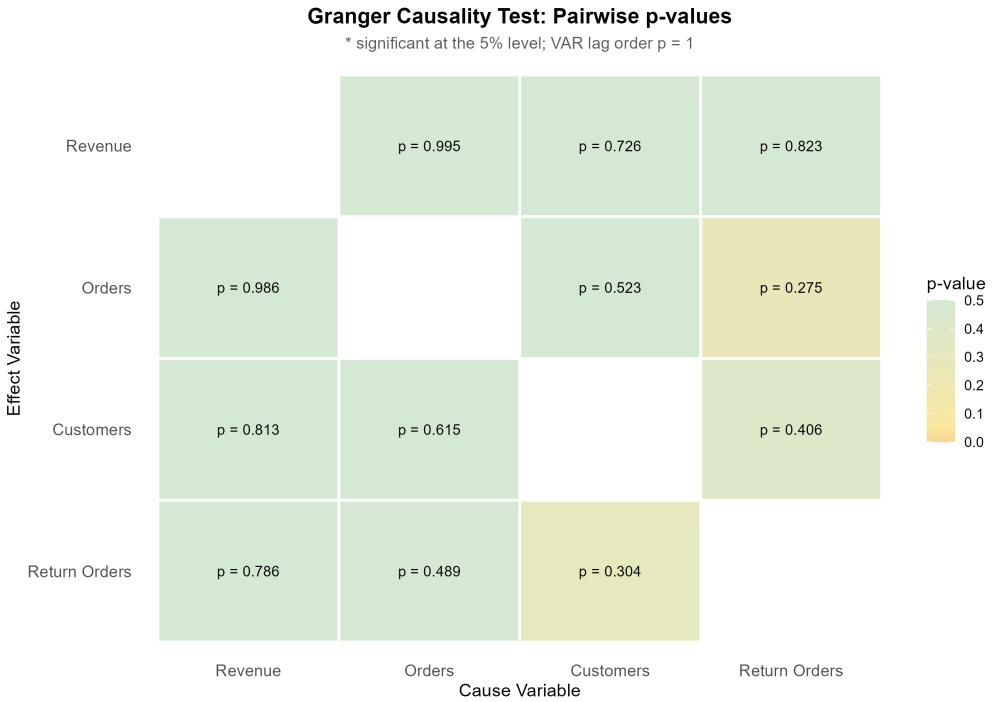


Figure 10: Pairwise Granger causality test p-values for the VAR(1) system.

5.2.3 Impulse Response Results

Figure 11 shows the response of $\Delta \log(\text{Weekly Revenue})$ to one-standard-deviation orthogonalised shocks in orders, customers, and return orders. The horizon is 12 weeks, with 95% bootstrap confidence intervals.

The order shock produces a positive first-week response, peaking near 0.026. However, the confidence interval includes zero. The response turns slightly negative in the second week and then fades quickly. The customer shock is smaller, peaking around 0.010, and is also statistically uncertain. The return-order shock stays close to zero throughout the horizon. Overall, cross-variable shocks have weak and short-lived effects on revenue growth.

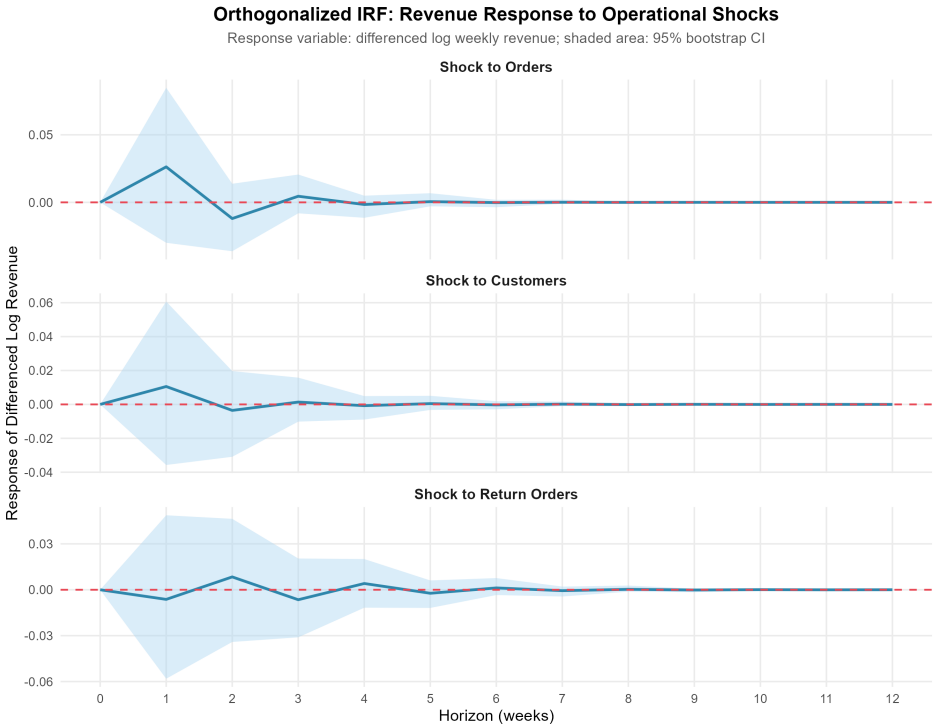


Figure 11: Orthogonalised impulse response functions: response of $\Delta \log(\text{Weekly Revenue})$ to one-standard-deviation shocks in orders, customers, and return orders over a 12-week horizon.

5.2.4 Forecast Error Variance Decomposition

The FEVD gives the same message. Revenue's own innovations explain about 99% of its forecast error variance across the 12-week horizon. Order shocks account for roughly 1%, while customer and return-order shocks contribute little.

After weekly aggregation and first differencing, short-run revenue uncertainty is therefore driven mainly by revenue's own movements. Orders, customers, and return orders add little incremental information in this VAR specification.

5.2.5 Overall Interpretation

The VAR(1) results give a cautious dynamic picture. The system is stable and passes the residual serial-correlation diagnostic, but the pairwise Granger tests do not find significant predictive directions at the 5% level. Orders, customers, and return orders do not improve one-week-ahead prediction of revenue growth. Revenue growth also does not predict next-week return-order growth.

The impulse responses support this conclusion. Order and customer shocks generate small first-week

revenue responses, but the confidence intervals include zero. Return-order shocks have no clear revenue effect. The FEVD further shows revenue forecast uncertainty is almost entirely explained by revenue's own shocks.

These results do not imply that orders, customers, or returns are operationally unimportant. They show a narrower point: after differencing, weekly aggregation, and conditioning on the full VAR(1), lagged cross-variable signals are weak. This complements the exploratory and return-lag analyses. Contemporaneous level relationships and medium-horizon growth co-movement may still matter, even when the one-lag VAR evidence is statistically conservative.

5.3 Experiment III: Return Lag and Abnormal Sales Peak Results

Using the threshold $\tau = 2$ ($Z_t^S > 2$), four abnormal revenue peaks are detected, mainly in late 2010 and late 2011. The largest occurs on 2011-12-05, with weekly revenue of 503,785.75 and a z-score of 3.83. Under the same threshold, three abnormal return-order peaks are found on 2010-02-01, 2010-05-10, and 2010-11-01. The largest reaches 314 orders, with a z-score of 4.23.

The peak-matching test gives a null result. None of the four abnormal revenue peaks is followed by an abnormal return-order peak within four weeks. This may reflect the different seasonal timing of the two peak types. Revenue peaks cluster near year-end, while return-order peaks occur in other months. The small number of revenue peaks also limits power. Thus, the null result should be read as inconclusive, not as evidence against a return-lag mechanism.

Table 12 reports the forward correlations $\widehat{\rho}_{SB}(k)$ for $k = 0, 1, 2, 3, 4$. The strongest associations appear at the shortest horizons. The contemporaneous correlation is significant at $\widehat{\rho}_{SB}(0) = 0.387$ ($p < 0.001$), and the one-week-ahead correlation is also significant at $\widehat{\rho}_{SB}(1) = 0.331$ ($p < 0.001$). Lag 2 is borderline ($\widehat{\rho}_{SB}(2) = 0.185$, $p = 0.060$), while lags 3 and 4 are not significant. The level-based revenue–return association is therefore concentrated in the current week and the following week.

Table 12: Forward Pearson correlations between weekly revenue S_t and future weekly return orders B_{t+k}, for $k = 0$ to 4. p-values are from two-sided t-tests; n is the number of observation pairs. *** p<0.001.

Lag k (weeks)	$\widehat{\rho}_{SB}(k)$	p-value	n
0	0.387^{***}	< 0.001	106
1	0.331^{***}	< 0.001	105
2	0.185	0.060	104
3	0.143	0.149	103
4	0.025	0.807	102

These level-based results should be read alongside the differenced-log evidence in Sections 3.3 and 5.2. The forward correlations measure whether high-revenue weeks are followed by high return volumes. By contrast, the CCF and VAR analyses focus on growth-rate dynamics after differencing. The two views are complementary. In levels, return pressure appears quickly, within the same week or the next week. In growth rates, the delayed response may extend toward the four-week horizon.

Operationally, return monitoring should increase immediately after high-revenue weeks. The main response window is the current and next week. A longer four-week watch window is still useful after major demand peaks, given the delayed growth-rate evidence from the CCF and the broader dynamic checks from the VAR analysis.

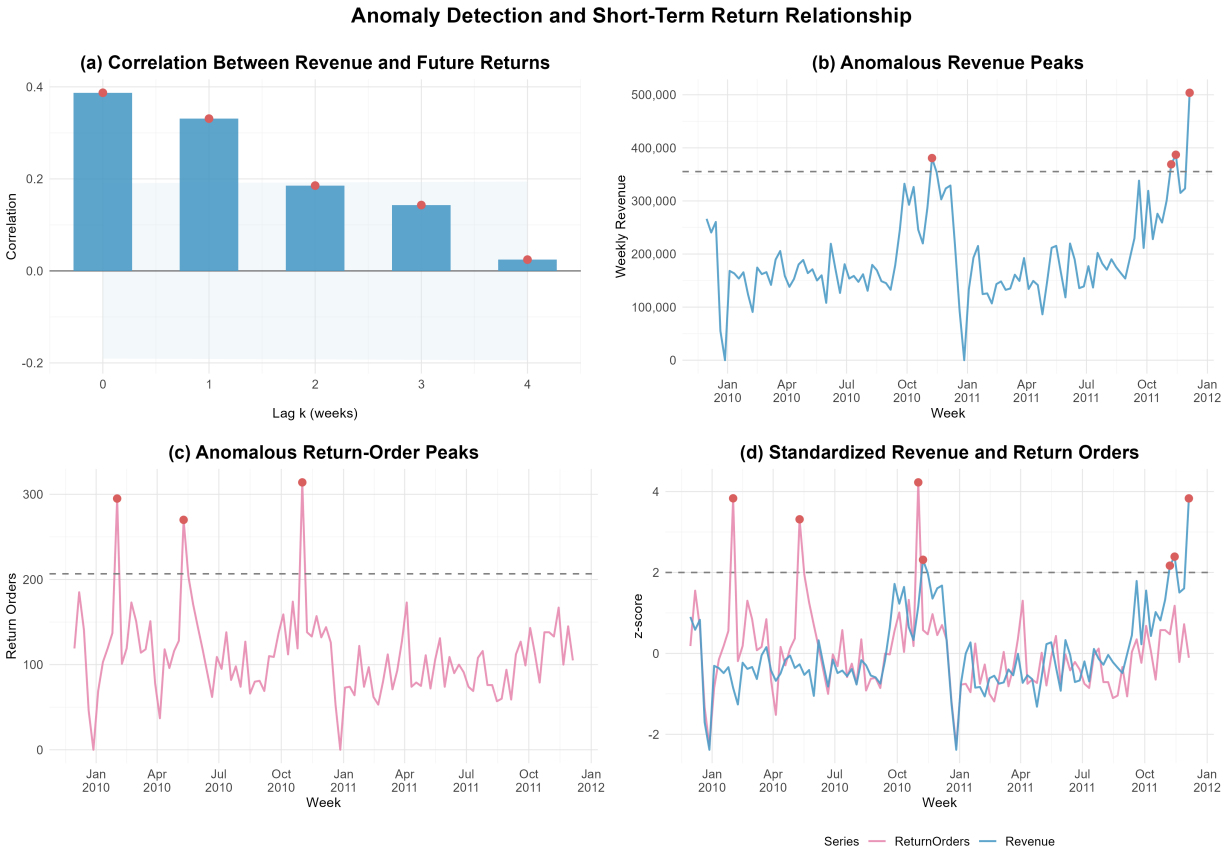


Figure 12: Anomaly detection and short-term revenue–return relationship. Panels show: (a) weekly revenue with abnormal peaks ($Z_t^S > 2$); (b) weekly return/cancellation orders with abnormal peaks ($Z_t^B > 2$); (c) standardized z-score overlay of both series; and (d) forward Pearson correlations $\widehat{\rho}_{SB}(k)$ for $k = 0$ to 4, with 95% confidence bounds.

6 Discussion

6.1 Model Selection Logic and Performance Evaluation

The forecasting models form a progression from seasonal benchmarks to increasingly structured specifications. They follow a simple progression from benchmarks to more structured specifications. Seasonal Naive is first used as a basic but meaningful benchmark, because the exploratory analysis shows a strong weekly pattern in daily revenue. Its competitive performance confirms that weekly recurrence is one of the most stable signals in the data. By contrast, the non-seasonal ARIMA model performs poorly because it does not explicitly account for the seven-day cycle. This comparison shows that adding model complexity is not useful unless the main seasonal structure is correctly specified.

SARIMA improves on ARIMA by directly modeling weekly seasonality through seasonal differencing and seasonal ARMA terms. Its stronger performance, especially in the year-end hold-out period, suggests that stochastic seasonal dynamics are more suitable for this revenue series than a purely non-seasonal specification. In this sense, SARIMA is not selected simply because it is more complex, but because its structure matches the main empirical pattern observed in the data.

Prophet and DHR-ARIMA provide a different way to model seasonality. Instead of treating seasonality as stochastic dependence, they represent trend and seasonal components deterministically. These models

perform well in some rolling validation periods, which suggests that deterministic trend-seasonality can be useful when the demand pattern is relatively smooth. However, their advantage weakens in the final year-end hold-out window. This result is plausible given the short sample and the sharp seasonal surge near the forecast boundary. With limited annual cycles, deterministic yearly components may be less stable when extrapolated into a volatile peak period.

SARIMAX extends the forecasting framework by adding operational predictors. The results suggest that not all external variables are equally useful. Calendar and event-window indicators add little beyond the seasonal structure already captured by SARIMA and the weekly benchmark. Lagged operational variables are more informative, because they reflect recent order and return/cancellation activity available before the forecast date. Even so, their improvement is limited. This indicates that operational covariates help explain part of the remaining short-run dependence, but they do not replace the central role of historical revenue and weekly seasonality.

Overall, the forecasting results support a cautious model selection strategy. Weekly seasonality is the primary signal, Seasonal Naive is a strong benchmark, and SARIMA provides the most reliable extension when stochastic variation is important. Prophet and DHR-ARIMA are useful robustness comparisons rather than consistently dominant models. Lagged-only SARIMAX adds operational information, but its role is supplementary. Therefore, Rolling SARIMA is the most reasonable primary forecasting model for this dataset, while Seasonal Naive should be retained as a benchmark and lagged operational regressors can be used as a diagnostic or supplementary extension.

6.2 Managerial Implications

For demand planners, the main forecasting signal is weekly seasonality together with year-end demand concentration. This suggests that inventory planning should not rely only on year-over-year extrapolation, which may underreact to the sharp late-year increase shown in Figure 1. Rolling SARIMA forecasts provide a more useful operational reference because they update the information set and capture the weekly demand pattern. The Monday-to-Thursday workday pattern also suggests that replenishment can be concentrated on high-demand weekdays rather than spread uniformly across the week.

For logistics coordinators, the strong contemporaneous correlation between orders and customers suggests that customer activity can serve as a rough same-week indicator of order volume. However, the VAR results do not provide strong one-week-ahead predictive evidence, so this signal should be used for monitoring rather than as a standalone forecasting rule. In practice, logistics teams may combine rolling revenue forecasts with same-week order and customer indicators to adjust fulfillment capacity more cautiously.

For after-sales teams, the return-lag results suggest that return/cancellation pressure is mainly concentrated in the revenue-spike week and the following week. A short response window around weeks 0–1 is therefore operationally important, while a longer four-week monitoring window can be retained as a cautious follow-up based on the CCF evidence. For cross-border platforms, this extended monitoring window is still useful because international transit and customs processing may delay return flows. The seasonal peaks in return rates during January, February, and May in Figure 4 further suggest that after-sales resources should be prepared before these periods rather than only after return volumes rise.

6.3 Research Limitations

The two-year sample provides only two complete annual cycles, which limits the reliability of long-run trend inference and annual Fourier parameter estimates. Prophet's instability at the year-end evaluation boundary is partly a consequence of this constraint. A longer history with more annual cycles would provide more stable estimates of annual seasonality. The VAR system, estimated on approximately 100 usable weekly observations, is adequate for a VAR(1) with four variables but cannot support higher-lag or higher-dimensional specifications, restricting the model's ability to capture longer-horizon dynamic interactions.

Several important demand drivers are absent from the dataset, including promotional campaigns, advertising expenditure, platform traffic, and holiday marketing schedules. The failure of calendar-based SARIMAX specifications to improve forecast accuracy may therefore reflect a data gap rather than the operational irrelevance of marketing activity. In cross-border e-commerce, sales peaks are often tied to platform-specific promotional events that standard calendar indicators cannot adequately represent, introducing an unknown but potentially substantial source of unexplained revenue variance. Additionally, the non-random pattern of missing Customer ID values introduces measurement error into the customer count series and may attenuate the estimated Granger coefficients; this does not affect the revenue or order series and leaves the primary forecasting results intact, but warrants attention in any replication using customer count as a predictor.

Finally, the VAR framework assumes linearity and parameter stability throughout the sample. Return dynamics are plausibly asymmetric—the return response to a revenue surge may be proportionally larger than the reduction in return pressure following a decline—and may shift structurally around major demand events such as the 2011 year-end spike. These nonlinearities are not captured by the estimated VAR(1) and represent a natural direction for methodological refinement in future work.

7 Conclusion

Based on transaction-level records from a cross-border e-commerce platform, this study constructs daily and weekly time series to address three linked questions: daily revenue forecasting, weekly multivariate dynamics, and return/cancellation behavior following high-sales periods. Rather than benchmarking models in isolation, the analysis connects these three perspectives within a common empirical framework—a design that better reflects the operational complexity of retail forecasting.

Across the forecasting experiments, model choice matters less than structural alignment with the data. Weekly seasonality is the dominant and most stable signal; Seasonal Naive therefore remains a meaningful benchmark, while SARIMA extends it by capturing stochastic seasonal dependence. Prophet and DHR-ARIMA perform well in smoother validation windows but lose their edge near the volatile year-end hold-out. Lagged operational covariates add short-run information, though their contribution is modest. On the multivariate side, the VAR results are conservative: contemporaneous correlations among revenue, orders, customers, and return/cancellation orders are strong, but one-step-ahead Granger predictability is limited, and revenue forecast errors are driven mainly by revenue's own shocks. The return-lag analysis further shows that post-peak return pressure concentrates in the spike week and the one immediately following, with only weak evidence of longer delays.

Practically, these findings suggest that retailers should maintain simple seasonal benchmarks, use SARIMA-type models as the primary operational forecasting model, and treat operational predictors as diagnostic supplements rather than substitutes for revenue history. Future work could strengthen this framework by extending the sample to more annual cycles, introducing segment-level modeling at the product or country level, and evaluating probabilistic and machine-learning methods under the same rolling-origin design. Nonlinear or regime-switching specifications may also prove useful for examining whether revenue–return dynamics differ between normal and peak-demand periods.

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Division of Work

Table 13: Division of Work by Author

Task	Lv Bing	He Jiali
Individual work		
§3.3–3.4 Return/cancellation analysis; stationarity tests; ACF/PACF	✓	
§4.2–4.3 VAR and return-lag methodology	✓	
§5.2–5.3 VAR results; Granger causality; IRF/FEVD; anomaly analysis	✓	
§2 cleaning and time series construction		✓
§3.1–3.2 EDA: trend, seasonality, and operational variables		✓
§4.1 Forecasting methodology		✓
§5.1 Forecasting results and model comparison		✓
Appendix Variable description and model settings		✓
Joint work		
Topic selection, research design, and data acquisition	✓	✓
Literature review	✓	✓
§1, §6, §7 Introduction, Discussion, and Conclusion	✓	✓
Abstract and overall revision	✓	✓

A Data Description and Cleaning Records

This appendix provides supplementary information on the raw variables, descriptive statistics, missing values, and cleaning records. Table 14 reports the original variables in the Online Retail II dataset. Additional cleaning records and sample-size comparisons are included to facilitate replication.

Table 14: Description of Original Variables

Variable	Type	Description
Invoice	String	Invoice or order number; entries beginning with C indicate cancellations.
StockCode	String	Product code.
Description	String	Product description.
Quantity	Numeric	Number of units in the transaction line.
InvoiceDate	Timestamp	Date and time of the transaction.
Price	Numeric	Unit price of the product.
Customer ID	Identifier	Customer identifier, with many missing values.
Country	String	Customer's country of residence.

B Model Diagnostics and Supplementary Results

This appendix reports model implementation settings and supplementary diagnostic figures for the forecasting, VAR, and return-lag analyses.

B.1 Model Implementation and Parameter Selection

Table 15 summarizes the implementation details and parameter selection rules for the daily forecasting models. Table 16 reports the corresponding settings for the weekly VAR and return-lag analyses.

Table 15: Implementation Details and Parameter Selection for Daily Forecasting Models

Model / Setting	Implementation	Parameters Used	Selection Rule / Notes
Naive	`forecast::naive()` on $X_t = \log(1 + R_t)$	Frequency $s = 7$; final hold-out horizon $H = 56$	Fixed benchmark; no tuning. Forecasts are back-transformed by $\exp(\widehat{X}_t) - 1$ and truncated at zero if negative.
Seasonal Naive	`forecast::snaive()` on X_t	Weekly seasonality $s = 7$; final hold-out horizon $H = 56$	Fixed benchmark using the previous value from the same weekday.
ARIMA, fixed-origin	`auto.arima` with `seasonal = FALSE`	Selected model: ARIMA$(0, 0, 5)$ with non-zero mean	Orders selected on the training sample. Full search is used first, with stepwise fallback if needed.
SARIMA, fixed-origin	`auto.arima` with `seasonal = TRUE` and weekly period $s = 7$	Selected model: ARIMA$(3, 0, 0)(2, 1, 0)_7$	Orders selected on the training sample. Seasonal differencing is imposed first, with fallback search if needed.
Prophet full, fixed-origin	Prophet fitted to X_t	Weekly seasonality = TRUE; yearly seasonality = TRUE; daily seasonality = FALSE; additive specification; 95% interval	Explicitly fixed Prophet full specification. Code contains a weekly-only fallback if full Prophet fails, but the final paper tables use Prophet full.
DHR-ARIMA, fixed-origin	Dynamic harmonic regression with ARIMA errors	Weekly Fourier order $K_w = 3$; annual Fourier candidates $K_y = 1, \ldots, 4$; selected $K_y = 4$; selected ARIMA error: ARIMA$(3, 0, 4)$	K_y selected by training-sample AICc; ARIMA error selected by `auto.arima`.
SARIMAX, fixed-origin	Regression with seasonal ARIMA errors and external regressors	Selected error model: ARIMA$(1, 0, 3)(0, 0, 2)_7$. Regressors include weekday, month, December indicator, weekend indicator, lag-1 and lag-7 orders, and lag-1 and lag-7 return/cancellation orders.	Error structure selected by `auto.arima` with `xreg`. Same-day operational variables are excluded. Fixed-origin multi-step SARIMAX should be interpreted cautiously because some lagged regressors inside the hold-out period may not be available at the initial forecast origin.
Rolling Seasonal Naive	One-step-ahead rolling forecast	Uses X_{t-7}	Fixed benchmark. The information set uses observations before the forecast date.
Rolling SARIMA	Refit at each one-step-ahead forecast origin	Fixed ARIMA$(3, 0, 0)(2, 1, 0)_7$ structure	The fixed order is taken from the fixed-origin SARIMA benchmark; model is refitted using an expanding window.
Rolling Prophet full	Refit Prophet at each one-step-ahead forecast origin	Weekly seasonality = TRUE; yearly seasonality = TRUE; daily seasonality = FALSE; additive specification	Fixed Prophet full specification; fitted using only observations before each forecast date.

Model / Setting	Implementation	Parameters Used	Selection Rule / Notes
Rolling Lagged-Only SARIMAX	Refit SARIMAX at each one-step-ahead forecast origin	Fixed ARIMA$(3,0,0)(2,1,0)_7$ error structure; regressors include lag-1 and lag-7 orders and return/cancellation orders	Fixed error order; expanding-window refit. Same-day operational variables are excluded, so this is the main SARIMAX setting for realistic one-step-ahead interpretation.
Rolling-origin CV	Expanding-window one-step-ahead cross-validation	CV period: 2011-06-01 to 2011-10-14; 136 forecasts; final 56-day hold-out excluded	Used as a robustness check before the final hold-out period. Rolling SARIMA, SARIMAX, and Prophet full follow the corresponding rolling specifications above.

Table 16: Implementation Details for VAR and Return-Lag Analyses

Module / Method	Implementation	Parameters Used	Selection Rule / Notes
Weekly VAR variables	VAR estimated on transformed weekly variables	Variables: $\Delta\log(1 + Revenue_t)$, $\Delta\log(1 + Orders_t)$, $\Delta\log(1 + Customers_t)$, and $\Delta\log(1 + ReturnOrders_t)$	First and last incomplete weeks are removed. Christmas–New Year shutdown weeks with large log-differences are also excluded.
VAR sample	Weekly differenced-log system	Final sample: $n = 99$, from 2009-12-14 to 2011-11-28	Used for lag selection, diagnostics, Granger tests, IRFs, and FEVD.
VAR lag selection	`VARselect` with constant term	Candidate maximum lag: 8; final lag order: $p = 1$	AIC, HQ, SC/BIC, and FPE are computed; SC/BIC is prioritized due to the limited weekly sample.
VAR diagnostics	Stability and residual serial-correlation checks	Characteristic roots have modulus below one; Portmanteau test uses lag 12	The fitted VAR is stable; Portmanteau p-value is 0.3186 in the code output.
Granger causality	Pairwise restricted/unrestricted F-tests	All ordered pairs are tested under the selected VAR lag $p = 1$	Implemented as pairwise tests rather than system-level causality tests.
Impulse response functions	Orthogonalized IRFs	Horizon = 12 weeks; bootstrap = TRUE; confidence level = 95%; bootstrap runs = 200	Cholesky ordering follows $Revenue \rightarrow Orders \rightarrow Customers \rightarrow ReturnOrders$.
Forecast error variance decomposition	Orthogonalized FEVD	Horizon = 12 weeks	Uses the same Cholesky ordering as the IRF analysis.
Forward return correlation	Pearson correlation between current weekly revenue and future return/-cancellation orders	Raw weekly levels; lags $k = 0, \ldots, 4$	Used to check whether high-revenue weeks are followed by higher return/cancellation volumes.

Module / Method	Implementation	Parameters Used	Selection Rule / Notes
Abnormal peak matching	Z-score based peak detection	Revenue and return/cancellation order peaks are defined by $z > 2$; future return/cancellation peaks are checked within $t+1, \ldots, t+4$	Same-week matching is excluded in the peak-lag analysis.

B.2 Supplementary Diagnostic Figures and Tables

This subsection reports additional diagnostic figures not included in the main text. The Prophet comparison is included to show the robustness check behind the final choice of the full Prophet specification. The remaining figures summarize residual autocorrelation patterns for the forecasting models under the three evaluation schemes.

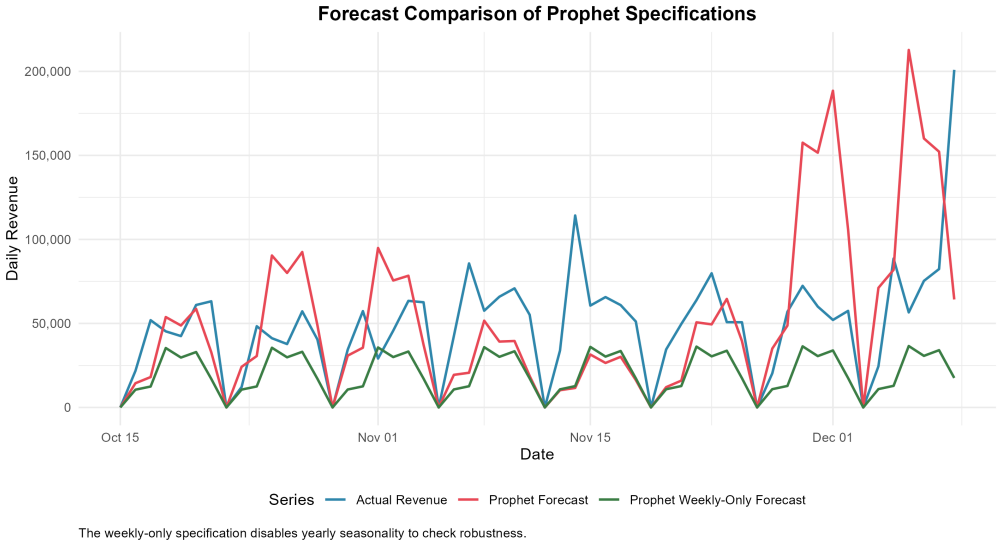


Figure 13: Forecast comparison of Prophet specifications.

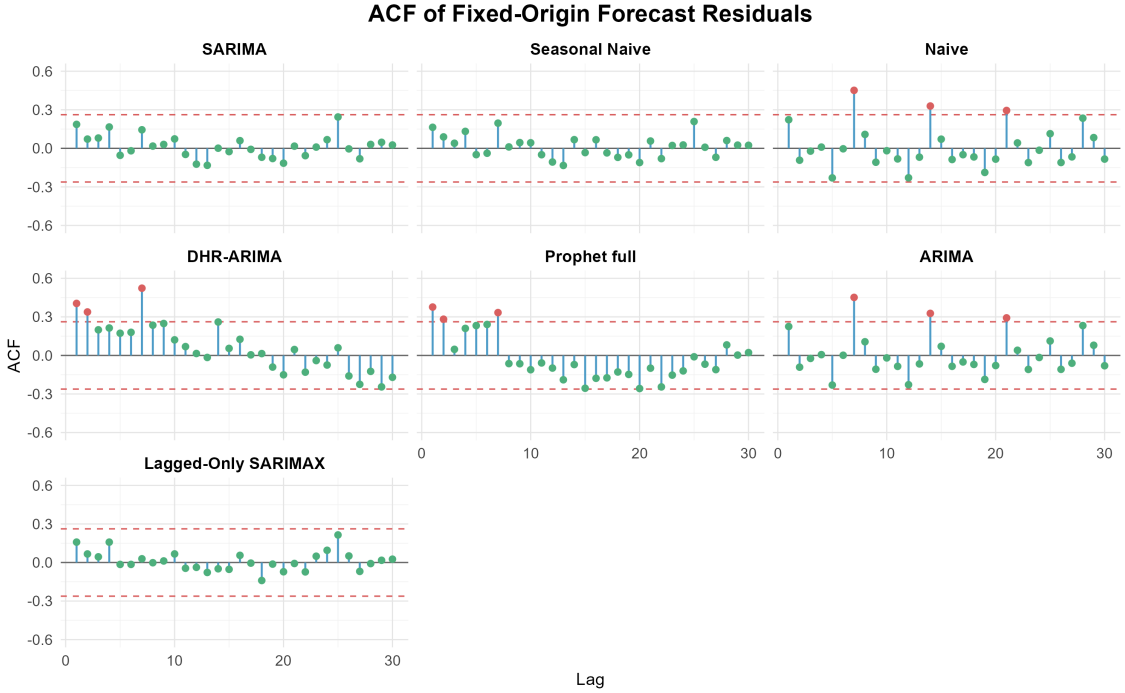


Figure 14: ACF of fixed-origin forecast residuals across models.

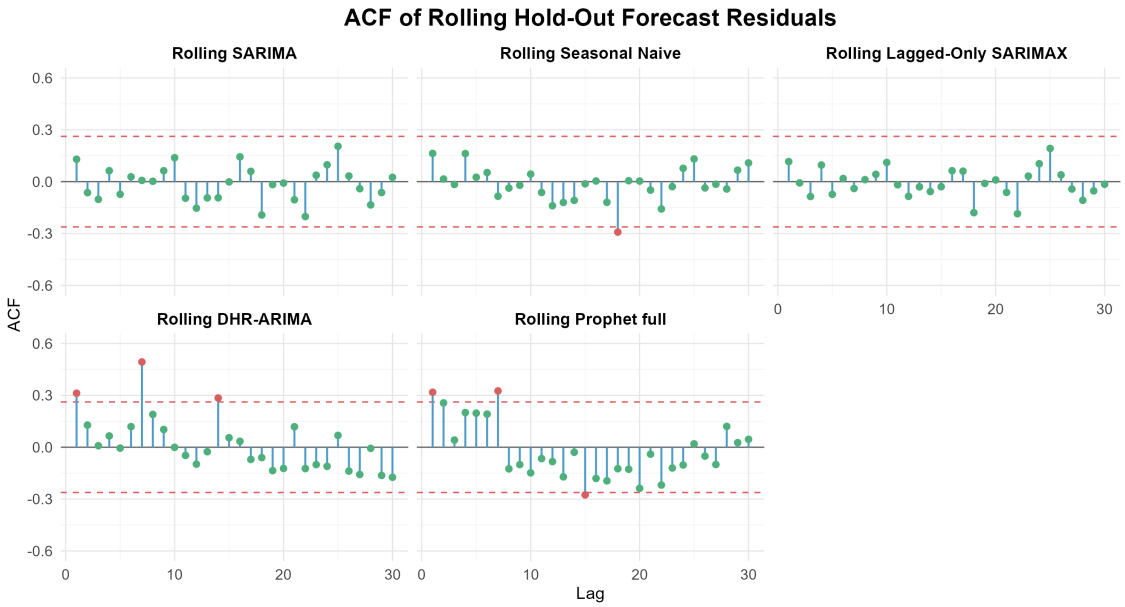


Figure 15: ACF of rolling hold-out forecast residuals across models.

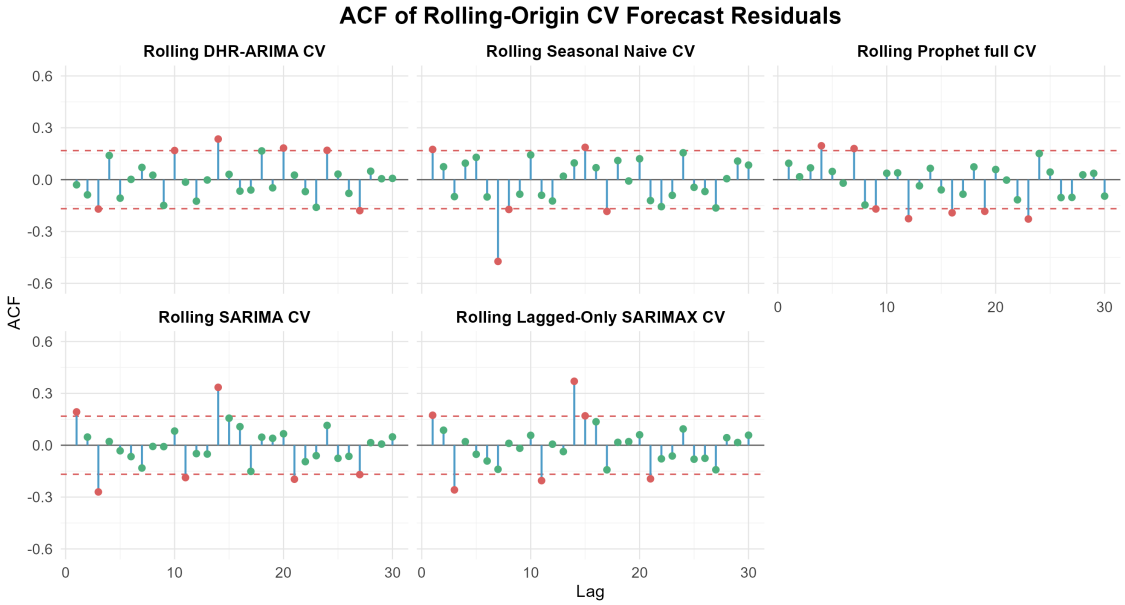


Figure 16: ACF of rolling-origin cross-validation forecast residuals across models.

C Code Implementation

This appendix gives abridged but faithful R code excerpts from the empirical-analysis scripts. The excerpts retain core time-series construction, target transformations, forecasting models, evaluation designs, residual diagnostics, weekly VAR analysis, and return-lag analysis. Upstream record cleaning, file output, figure formatting, and repeated data-wrangling checks are omitted for brevity.

C.1 Time-Series Construction

The daily and weekly series are constructed from cleaned transaction records. Normal sales define revenue, order, and customer measures, while cancellation or return records define return-order measures. Missing calendar dates and weeks are filled with zero before log transformations and weekly first differences

are applied.

```r
# Construct daily operational series
daily_sales <- sales_clean %>%
  group_by(Date) %>%
  summarise(
    revenue = sum(Revenue, na.rm = TRUE),
    orders = n_distinct(Invoice),
    customers = n_distinct(CustomerID),
    .groups = "drop"
  )

daily_returns <- returns_clean %>%
  group_by(Date) %>%
  summarise(return_orders = n_distinct(Invoice), .groups = "drop")

daily_retail_ts <- full_dates %>%
  left_join(daily_sales, by = "Date") %>%
  left_join(daily_returns, by = "Date") %>%
  mutate(across(where(is.numeric), ~ replace_na(.x, 0))) %>%
  mutate(
    log_revenue = log1p(revenue),
    log_orders = log1p(orders),
    log_customers = log1p(customers),
    log_return_orders = log1p(return_orders)
  )

# Construct weekly variables for dynamic analysis
weekly_retail_ts <- full_weeks %>%
  left_join(weekly_sales, by = "Week") %>%
  left_join(weekly_returns, by = "Week") %>%
  mutate(across(where(is.numeric), ~ replace_na(.x, 0))) %>%
  mutate(
    log_weekly_revenue = log1p(weekly_revenue),
    log_weekly_orders = log1p(weekly_orders),
    log_weekly_customers = log1p(weekly_customers),
    log_weekly_return_orders = log1p(weekly_return_orders)
  ) %>%
  arrange(Week) %>%
  mutate(
    d_log_weekly_revenue = log_weekly_revenue - lag(log_weekly_revenue),
    d_log_weekly_orders = log_weekly_orders - lag(log_weekly_orders),
    d_log_weekly_customers = log_weekly_customers - lag(log_weekly_customers),
    d_log_weekly_return_orders =
      log_weekly_return_orders - lag(log_weekly_return_orders)
  )
```

C.2 Forecasting Target and Accuracy Metrics

All daily forecasting models use `log1p(revenue)` as the target. Forecasts are transformed back to the original revenue scale with `expm1`, with negative values truncated at zero. Accuracy is evaluated using MAE, RMSE, and sMAPE on the original revenue scale.

```r
# Forecasting target and inverse transformation
daily_model_data <- daily_model_data %>%
  mutate(log_revenue = log1p(revenue))

forecast_revenue <- pmax(expm1(forecast_log_revenue), 0)

```

```r
# Accuracy metrics on the original revenue scale
smape <- function(actual, forecast_value) {
  denominator <- abs(actual) + abs(forecast_value)
  terms <- if_else(
    denominator == 0,
    0,
    2 * abs(actual - forecast_value) / denominator
  )
  mean(terms, na.rm = TRUE) * 100
}

calc_metrics <- function(forecast_table) {
  forecast_table %>%
    group_by(model) %>%
    summarise(
      MAE = mean(abs(actual_revenue - forecast_revenue), na.rm = TRUE),
      RMSE = sqrt(mean((actual_revenue - forecast_revenue)^2, na.rm = TRUE)),
      sMAPE = smape(actual_revenue, forecast_revenue),
      .groups = "drop"
    )
}
```

C.3 Fixed-Origin Forecasting

The fixed-origin evaluation uses the final 56 daily observations as the hold-out set. Baseline, ARIMA, SARIMA, Prophet full, and DHR–ARIMA models are estimated from the same training sample.

```r
# Fixed-origin train/test split
make_train_test_split <- function(data, test_days = 56) {
  data <- data %>% arrange(date)
  list(
    train = data %>% slice_head(n = nrow(data) - test_days),
    test = data %>% slice_tail(n = test_days)
  )
}

split_data <- make_train_test_split(daily_model_data, test_days = 56)
train_data <- split_data$train
test_data <- split_data$test
horizon <- nrow(test_data)

# Fixed-origin baselines, ARIMA, and SARIMA
train_ts <- ts(train_data$log_revenue, frequency = 7)

naive_forecast <- forecast::naive(train_ts, h = horizon)
snaive_forecast <- forecast::snaive(train_ts, h = horizon)

arima_model <- forecast::auto.arima(
  train_ts,
  seasonal = FALSE,
  stepwise = FALSE,
  approximation = FALSE
)

sarima_model <- forecast::auto.arima(
  train_ts,
  seasonal = TRUE,
  D = 1,
  stepwise = FALSE,
```

```
  approximation = FALSE
)

# Prophet full specification
prophet_model <- prophet::prophet(
  train_data %>% transmute(ds = date, y = log_revenue),
  yearly.seasonality = TRUE,
  weekly.seasonality = TRUE,
  daily.seasonality = FALSE,
  seasonality.mode = "additive",
  interval.width = 0.95
)

# DHR-ARIMA Fourier regressors and K selection
make_fourier_terms <- function(data, K_week = 3, K_year = 1) {
  t <- as.integer(data$date - min(data$date)) + 1
  weekly <- forecast::fourier(ts(rep(0, length(t)), frequency = 7), K = K_week)
  annual <- forecast::fourier(ts(rep(0, length(t)), frequency = 365.25), K = K_year)
  cbind(weekly, annual)
}

fit_dhr_arima <- function(train_data, K_week = 3, K_year = 1) {
  forecast::auto.arima(
    y = ts(train_data$log_revenue),
    xreg = make_fourier_terms(train_data, K_week, K_year),
    seasonal = FALSE,
    stepwise = TRUE,
    approximation = FALSE
  )
}

K_week <- 3
K_year_candidates <- 1:4
selected_K_year <- K_year_candidates[
  which.min(sapply(K_year_candidates, function(K) {
    fit_dhr_arima(train_data, K_week, K)$aicc
  }))
]
```

C.4 Rolling Forecasting and SARIMAX

The rolling hold-out design is one-step-ahead: for each forecast date, models are fitted using only earlier observations. Rolling SARIMA and lagged-only SARIMAX use a fixed ARIMA$(3,0,0)(2,1,0)_7$ error structure, while Prophet is refitted with the full additive specification.

```
# One-step-ahead rolling information set
for (current_date in test_data$date) {
  history_data <- all_data %>%
    filter(date < current_date) %>%
    arrange(date)

  actual_data <- test_data %>% filter(date == current_date)

  # Refit rolling models at this forecast origin.
}

# Rolling SARIMA error structure
sarima_fit <- forecast::Arima(
```

```
  history_ts,
  order = c(3, 0, 0),
  seasonal = list(order = c(2, 1, 0), period = 7),
  include.constant = FALSE,
  method = "ML"
)

# Rolling Lagged-Only SARIMAX regressors
train_xreg <- history_data %>%
  arrange(date) %>%
  mutate(
    lag1_orders = lag(orders, 1),
    lag7_orders = lag(orders, 7),
    lag1_return_orders = lag(return_orders, 1),
    lag7_return_orders = lag(return_orders, 7)
  ) %>%
  select(lag1_orders, lag7_orders,
        lag1_return_orders, lag7_return_orders)

sarimax_fit <- forecast::Arima(
  history_ts,
  order = c(3, 0, 0),
  seasonal = list(order = c(2, 1, 0), period = 7),
  xreg = train_xreg,
  include.constant = FALSE,
  method = "ML"
)

# Rolling Prophet full
prophet_fit <- prophet::prophet(
  history_data %>% transmute(ds = date, y = log_revenue),
  yearly.seasonality = TRUE,
  weekly.seasonality = TRUE,
  daily.seasonality = FALSE,
  seasonality.mode = "additive",
  interval.width = 0.95
)

# Fixed-origin SARIMAX regressor candidates
calendar_xreg <- c("weekday", "month", "is_december", "is_weekend")
event_xreg <- c("is_christmas_window", "is_new_year_window")
lagged_xreg <- c(
  "lag1_orders", "lag7_orders",
  "lag1_return_orders", "lag7_return_orders"
)

# Same-day operational variables are excluded from rolling SARIMAX.
```

C.5 Rolling-Origin Cross-Validation

Rolling-origin cross-validation uses an expanding window and excludes the final 56-day hold-out period. The implemented validation range is 2011-06-01 to 2011-10-14 when these dates are available.

```
# Rolling-origin CV design
final_test_data <- daily_model_data %>% slice_tail(n = 56)
final_test_start <- min(final_test_data$date)

cv_start <- as.Date("2011-06-01")
cv_end <- as.Date("2011-10-14")
```

```
cv_data <- daily_model_data %>%
  filter(date >= cv_start, date <= cv_end, date < final_test_start)

for (current_date in cv_data$date) {
  history_data <- daily_model_data %>%
    filter(date < current_date) %>%
    arrange(date)

  actual_data <- cv_data %>% filter(date == current_date)

  # Models: Seasonal Naive CV, SARIMA CV, Prophet full CV,
  # Lagged-Only SARIMAX CV, and DHR-ARIMA CV.
}
```

C.6 Residual Diagnostics

Ljung–Box diagnostics are calculated from log-scale forecast residuals. The lag is set to $\min(14, \lfloor n/2 \rfloor)$, which is 14 for the 56-day fixed-origin and rolling hold-out evaluations.

```
# Log-scale residuals and Ljung-Box diagnostic
forecast_values <- forecast_values %>%
  mutate(residual_log = actual_log_revenue - forecast_log_revenue)

run_ljung_box <- function(forecast_table) {
  forecast_table %>%
    group_by(model) %>%
    summarise(
      lag = min(14, floor(n() / 2)),
      test = list(Box.test(residual_log, lag = lag[1],
                           type = "Ljung-Box")),
      statistic = as.numeric(test[[1]]$statistic),
      p_value = as.numeric(test[[1]]$p.value),
      .groups = "drop"
    ) %>%
    select(-test)
}
```

C.7 Weekly VAR Analysis

The VAR uses first differences of $\log(1+x)$ for weekly revenue, orders, customers, and return/cancellation orders. The final VAR lag is selected by SC/BIC from candidates up to lag 8. Stability, residual serial correlation, pairwise Granger tests, orthogonalized IRFs, and FEVD are then calculated from the fitted model.

```
# VAR input data
var_data <- weekly_retail_ts %>%
  transmute(
    Week,
    d_log_revenue = d_log_weekly_revenue,
    d_log_orders = d_log_weekly_orders,
    d_log_customers = d_log_weekly_customers,
    d_log_return_orders = d_log_weekly_return_orders
  ) %>%
  filter(complete.cases(.)) %>%
  slice(-1, -n()) %>%
```

```
  filter(
    abs(d_log_revenue) < 5,
    abs(d_log_orders) < 5,
    abs(d_log_customers) < 5,
    abs(d_log_return_orders) < 5
  )

# VAR lag selection and estimation
var_matrix <- var_data %>%
  select(d_log_revenue, d_log_orders,
         d_log_customers, d_log_return_orders)

lag_select <- vars::VARselect(var_matrix, lag.max = 8, type = "const")
selected_lag <- as.integer(lag_select$selection["SC(n)"])

var_model <- vars::VAR(var_matrix, p = selected_lag, type = "const")

# VAR stability and residual autocorrelation diagnostics
var_roots <- roots(var_model)
is_stable <- all(Mod(var_roots) < 1)

serial_test <- serial.test(
  var_model,
  lags.pt = 12,
  type = "PT.asymptotic"
)

# Pairwise Granger causality skeleton
lagged_data <- make_lagged_data(var_matrix, selected_lag)

restricted_model <- lm(effect ~ lags_except_cause, data = lagged_data)
unrestricted_model <- lm(effect ~ all_lag_terms, data = lagged_data)
granger_test <- anova(restricted_model, unrestricted_model)

# Orthogonalized IRF and FEVD
irf_results <- vars::irf(
  var_model,
  impulse = c("d_log_orders", "d_log_customers",
              "d_log_return_orders"),
  response = "d_log_revenue",
  n.ahead = 12,
  ortho = TRUE,
  boot = TRUE,
  ci = 0.95,
  runs = 200
)

fevd_results <- vars::fevd(var_model, n.ahead = 12)
```

C.8 Return-Lag and Abnormal Peak Analysis

The return-lag analysis uses raw weekly revenue and return/cancellation orders. It calculates forward correlations for lags $k = 0, \ldots, 4$ and identifies abnormal peaks using raw-level z-scores above 2. Each revenue peak is then checked for a return/cancellation peak within the following four weeks.

```
# Weekly raw-level return-lag variables
weekly_df <- raw_weekly %>%
  transmute(
```

```
    week = parse_date_safe(.data[[date_col]]),
    revenue = to_numeric_safe(.data[[revenue_col]]),
    return_orders = to_numeric_safe(.data[[return_col]])
  ) %>%
  group_by(week) %>%
  summarise(
    revenue = sum(revenue, na.rm = TRUE),
    return_orders = sum(return_orders, na.rm = TRUE),
    .groups = "drop"
  )

# Forward correlations for k = 0,...,4
max_lag_weeks <- 4

forward_correlation <- map_dfr(0:max_lag_weeks, function(k) {
  lag_df <- analysis_df %>%
    transmute(
      revenue_t = revenue,
      future_return_orders = lead(return_orders, k)
    ) %>%
    filter(complete.cases(.))

  tibble(
    lag_weeks = k,
    correlation = cor(lag_df$revenue_t,
                      lag_df$future_return_orders),
    p_value = cor.test(lag_df$revenue_t,
                       lag_df$future_return_orders)$p.value
  )
})

# Abnormal peak definition and matching rule
calc_z <- function(x) {
  (x - mean(x, na.rm = TRUE)) / sd(x, na.rm = TRUE)
}

z_threshold <- 2

analysis_df <- weekly_df %>%
  mutate(
    revenue_z_score = calc_z(revenue),
    return_z_score = calc_z(return_orders),
    is_revenue_peak = revenue_z_score > z_threshold,
    is_return_peak = return_z_score > z_threshold
  )

peak_matching <- revenue_anomalies %>%
  group_modify(function(.x, .y) {
    candidates <- analysis_df %>%
      filter(
        week > .x$revenue_peak_week[1],
        week <= .x$revenue_peak_week[1] + weeks(max_lag_weeks),
        is_return_peak
      )

    tibble(is_followed_by_return_peak = nrow(candidates) > 0)
  })
```